Divided for Good: Football Rivalries and Social Cohesion in Latin America

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July 7, 2026

Abstract

This paper examines the evolution of social cohesion in response to salient events that internally divide communities. The analysis focuses on football rivalries in Latin America, a fundamental cultural phenomenon in the region. Derbies induce polarization between the followers of both teams; however, since followers are not significantly segregated, derbies may also diminish the salience of other cleavages, such as socioeconomic status or ethnicity. By combining quasi-experimental variation in the timing of matches with public opinion surveys, I find that attitudes and beliefs conducive to social cohesion tend to improve for up to 30 days following a match. The effect is moderated by the behavior of football players, as captured by the number of red cards awarded during the game, suggesting that players-as-role-models is an important ingredient of such events.

JEL Codes: D74, D91, Z13, Z20
Keywords: social cohesion; football rivalries; role models; Latin America

*Universidad de los Andes, Chile. This paper benefited from comments by Dan Bjorkegren, Rob Blair, Ruben Durante, John Friedman, Brian Knight, Stelios Michalopoulos, Eduardo Montero, Bryce Steinberg, audience members at Brown and the 2024 ASREC Conference, and the advice of Pedro Dal Bó, for which I am very grateful. I also acknowledge financial support from ANID FONDECYT grant No. 11250280. E-mail: jpronconi@uandes.cl. Address: Av. Monseñor Álvaro del Portillo 12455, Santiago, Chile.

1 Introduction

Political polarization has surged in recent times, raising concerns about social divisions, conflict, and the future of democracy (Iyengar et al. 2019; Boxell, Gentzkow, and Shapiro 2024). An often mentioned cause is the behavior of many political leaders, who utilize and reinforce pre-existing cleavages as a political strategy, especially during electoral campaigns. These types of large-scale events that divide and polarize communities from within are conjectured to be critical for social cohesiveness but are understudied empirically, as well as the role played by the main actors involved in them.

In this paper, I address this issue by focusing on major football[1] rivalries in Latin America, a region where football is the most popular sport and is often considered a fundamental element of local cultures (Alabarces 2003). These events divide and polarize communities, potentially harming local cohesiveness. However, the divisions they induce transcend other cleavages, such as regionality, ethnicity, or socioeconomic status. Thus, while derbies antagonize individuals based on their football team identity, they have the potential to temporarily crowd out other, more divisive group identities and create opportunities for bonding on each side of the divide. Indeed, Social Identity Theory (Tajfel 1974) proposes that individuals may identify with multiple groups simultaneously, and that context is a key determinant of which group identity is most salient at each point in time (Shayo 2009).

I study the short-run impact of derbies on attitudes and beliefs conducive to social cohesion, by leveraging quasi-random variation arising from the coincidence of public opinion survey rounds with football matches. I combine a novel dataset of 104 derbies involving the 20 most important rivalries in Latin America with survey data from the AmericasBarometer, covering 11 countries between 2007 and 2019. I focus the analysis on a social cohesion index obtained as a function of two variables: trust in the local community and perceiving social conflict as the country's foremost problem. The former captures the proximate, horizontal dimension of trust toward the people nearby, while the latter reflects the perceived state of social relations at the societal scale. As such, the two variables tap distinct dimensions of social cohesion and, combined, provide a more comprehensive measure than either one alone. Moreover, since major public opinion surveys do not elicit football team preference, I use Google Trends data to construct regional-level measures of exposure to each rivalry in the sample.[2]

[1] I use the term "football" rather than "soccer" following the usual practice in the social sciences when studying this sport in the context of Latin America.

[2] Given the nature of the research design, the analysis of behavioral outcomes and long run effects is beyond the scope of this paper. However, a large body of work in economics has shown that survey data is

The baseline results indicate that social cohesion tends to *improve* in the aftermath of a derby, with the strongest effect taking place during the first 5 days (0.23 standard deviations) but remaining statistically significant up to a 30-day window (0.08 standard deviations). This result is in line with the idea that divisive events can paradoxically strengthen social cohesion by fostering unity among groups divided by deeper and more significant rifts. Crucially, however, the effect is strongly moderated by the behavior of players: As the number of red cards awarded during the game increases (reflecting violent or unethical behavior), the effect tends to reverse, becoming negative after games with 3 red cards or more. Effects are robust under each variable composing the social cohesion index separately as well as under a battery of checks, including alternative model specifications, testing for pre-trends, dropping one country at a time from the sample, reverse causality in the number of red cards, and adjusting standard errors for multiple hypothesis testing.

Besides red cards, I also explore how effects vary depending on whether the match ends in a draw or is of high stakes. There are no significant differences under the former, indicating that the existence of winners and losers does not affect the results. In contrast, effects tend to be stronger after high-stake matches, suggesting that the impact of the event grows with the amount of attention it receives. As for individual characteristics, I find no significant differences across gender, age, or educational attainment. This is consistent with the widespread interest in football across Latin America.

Finally, I examine the impact of derbies on additional outcomes, including perceived insecurity, general mood or optimism, and trust in the president. The results indicate no change in respondents' feelings of insecurity after a match, suggesting that changes in criminal activity are not a contributing factor. Additionally, respondents do not feel more optimistic about their country's or their own economic situation, implying that football matches do not create a general sense of well-being that could be influencing the baseline results. Lastly, there is no change in trust in the president, indicating that belief updating is limited to perceptions of fellow community members.

The findings point to novel policy opportunities for leveraging the social effects of major sporting events. Governments and civil society could time civic initiatives—such as public health or donation campaigns—to coincide with periods of heightened trust following local derbies. At the same time, related evidence showing that football can also be associated with conflict and violence in other settings (Card and Dahl 2011; Munyo and Rossi 2013; Bertoli 2017) suggests that these opportunities should be paired with efforts to preserve the

a source of meaningful information on individual behavior and social outcomes (Alesina and Giuliano 2015; Stantcheva 2023).

prosocial character of the event. Promoting fair play and peaceful conduct during these matches is therefore essential, as violent or unethical behavior can undermine their positive effects. These insights highlight the broader potential of emotionally resonant collective rituals to support policy efforts aimed at improving social cohesion.

This research relates to several strands of literature. First, to the large body of work that studies the determinants of interpersonal trust and social cohesion (Alesina and La Ferrara 2002; Nunn and Wantchekon 2011; Bazzi et al. 2019; Ronconi and Ramos-Toro 2025), especially the set of papers that considers the power of sports to generate intergroup contact and community-building (Depetris-Chauvin, Durante, and Campante 2020; Mousa 2020; Lowe 2021).[3]

Most closely related to my work, Depetris-Chauvin, Durante, and Campante (2020) explores how victories by national football teams in Africa promote social integration, as individuals tend to identify more with their nation than with their ethnic group, potentially leading to a reduction in civil conflict. My paper complements their study by focusing on a fundamentally different type of event within the realm of football: matches between local rivals, which divide communities from within along the lines of football identities, as opposed to uniting them against a foreign one. Moreover, I also complement their analysis by adding to the evidence that football players can act as role models for their communities. They observe more positive effects under more ethnically mixed team rosters; I find that effects tend to reverse as the number of red cards awarded increases, which reflects poor ethical behavior among players. Both sources of heterogeneity suggest role model effects. Additionally, Ajzenman, Ferman, and Sant'Anna (2023) presents results from a field experiment with fictitious Twitter profiles, where they interact different Brazilian political identities with football team identities, and show that highly polarized political identities can hinder the benefits to social interactions obtained from a shared football team identity. As in this paper, they leverage the fact that Latin American football rivalries tend to cut across other socioeconomic cleavages. Their result may imply that the effects documented in this paper are a lower bound, potentially being higher in less polarized contexts. In general, my work contributes to this literature by providing evidence that even highly polarized events at the intracommunity level have the potential to improve social cohesion, although subject to the proper and ethical behavior of the main characters involved.

Second, this paper speaks to the growing literature that documents the importance of

[3]Many other papers have studied sports as a window into understanding human behavior. See, for example, Palacios-Huerta (2003); Garicano, Palacios-Huerta, and Prendergast (2005); Card and Dahl (2011); Miguel, Saiegh, and Satyanath (2011); Schläpfer (2024); Glennon et al. (2025).

role models in determining beliefs and behavior (Jensen and Oster 2009; La Ferrara, Chong, and Duryea 2012), especially in developing countries (Lafortune, Riutort, and Tessada 2018; Dalton et al. 2021; Serra 2025). It demonstrates that football players, through their conduct on the field, can influence beliefs supporting social cohesion. This is consistent with the idea that players model how well-intentioned and peaceful interactions are possible, even with traditional rivals.

Finally, this research contributes to the set of papers studying the impact of mass sporting events on attitudes towards political incumbents (Healey, Malhotra, and Hyunjung Mo 2010; Fowler and Montagnes 2015; Busby, Druckman, and Fredendall 2017; Corbi 2018). Based on a large sample of respondents and quasi experimental variation, it shows that in Latin America—a region characterized by presidential systems—there is no evidence that football derbies have an effect on trust in the incumbent president.

2 Context

In the vast majority of Latin American countries, football is by far the most popular sport: around 75% of men and 45% of women are interested in it.[4] Far from being only a recreational activity, football represents a fundamental element of Latin American cultures (Alabarces 2003). For example, people are more likely to identify with a local football team than to be interested in the sport: almost 85% of men and 65% of women claim to support a local team.

An important feature of football in Latin America is that teams generally have a traditional rival—a team they have long confronted and toward which they came to develop a special sense of antagonism. Some famous examples include Boca Juniors v. River Plate in Argentina, Flamengo v. Vasco da Gama in Brazil, and Nacional v. Peñarol in Uruguay. Matches against the rival are generally the most important ones during the season and represent highly salient events.[5] These rivalries are known for stirring strong passions among the fans, sometimes leading to violence within, and in the area surrounding, football stadiums.

The other fundamental feature of Latin American football is that the followers of

[4]Based on Repucom (2014), NielsenSports (2018), and country-specific reports (COPES 2014; Opción Consultores 2014; Capital MX 2016; GfK Adimark 2016, 2017; Click 2018; Datafolha 2018; Datum Internacional 2018; GfK Adimark 2018, 2019; Ipsos 2019).

[5]Figure A1 in the appendix compares Google queries for some of the most important rivalries in Latin America in their respective countries to searches on the Superbowl in the United States. It shows spikes around the date of each match, with football rivalries being at least as salient as the Superbowl.

rival teams are not significantly segregated by characteristics such as location, ethnicity, or religion, unlike famous examples in other parts of the world, like Scotland's Celtics v. Rangers (Catholics v. Protestants), Spain's Barcelona v. Real Madrid (Catalans v. Madrileños), and, in US baseball, the Red Sox v. the Yankees (Boston v. New York residents). Instead, followers belong to the same communities and share their social networks, interacting with each other on a daily basis as friends, relatives, or neighbors. This implies that the cleavage induced by football rivalries tends to cut across other cleavages in the community, bringing together people from all socioeconomic groups, ages, genders, and religions behind the lines of each of the two teams involved.

Although it may be possible to find segregation among followers of smaller teams, this is not the case with big, first-division teams that have hundreds of thousands—if not millions—of followers, as the ones considered in this paper. Unfortunately, to the best of my knowledge, there are no data systematically documenting characteristics of football supporters in Latin America. However, I attempt to address this limitation in two ways. First, I leverage Google Trends data on the distribution of searches for each team across cities to provide suggestive evidence of limited geographical segregation (Figure A2). Second, I look into surveys collected by consulting firms in different countries, where they elicited football team preference and socioeconomic status of respondents. I find little differences in socioeconomic status between followers of rival teams. Figure A3 presents the evidence for Bolivian, Brazilian, Chilean, Ecuadorian, and Peruvian rivalries.

Finally, unlike other polarizing events (*e.g.*, political campaigns and elections), derbies do not impact other deeper determinants of social cohesion, such as expectations about policy changes or political incumbents. This makes football derbies a clean setting to study how divisive events affect social attitudes and allows to explore the impact of role models, captured by football players in this case.

3 Methods

In this section, I discuss the data and the empirical approach pursued to estimate the short-run, causal impact of football matches between rivals on attitudes and beliefs conducive to social cohesion.

3.1 Data

The analysis combines individual surveys, a novel dataset of football rivalries and matches, and Google search-based measures of rivalry salience across regions in Latin America.

3.1.1 Individual Attitudes

The data on individual attitudes and characteristics come from the AmericasBarometer, a public opinion survey conducted by the LAPOP. I include all waves between 2007 and 2019 collected in eleven Latin American countries: Argentina, Bolivia, Brazil, Chile, Colombia, Ecuador, Honduras, Mexico, Paraguay, Peru, and Uruguay.[6]

While the AmericasBarometer surveys a large range of attitudes, I focus on questions related to social cohesion and other potentially relevant outcomes. To measure social cohesion, I exploit two questions that were asked in all waves of the sample. One concerns interpersonal trust: "*Speaking of the people from around here, would you say that people in this community are very trustworthy, somewhat trustworthy, not very trustworthy or untrustworthy?*" The other question asks what the respondent perceives to be the most serious problem in the country: "*In your opinion, what is the most serious problem faced by the country?*" This is open ended, so responses are classified into thirty-seven topics by LAPOP, which I further classify into six broad categories: Society, Economy, Government, Infrastructure, Health & Education, and Other. Section A.2.1 in the appendix shows the mapping between topics and categories.

I combine the question on interpersonal trust with an indicator for perceiving social conflict as the country's foremost problem (that is, mentioning a problem in the Society category as the most serious one), obtaining a standardized index (Anderson 2008) that is the main outcome of interest in the paper (henceforth, the Social Cohesion Index). In particular, I build the index using all the information available in each response—that is, the full variability of the answers—so that interpersonal trust enters in its graded form rather than collapsed into a binary measure.

Social cohesion is widely understood as a multidimensional construct. Across the most influential frameworks, its core constituents are interpersonal trust, the quality of social relations (including tolerance and acceptance of diversity), a sense of belonging, and an orientation toward the common good and respect for shared social order (Chan, To, and Chan 2006; Dragolov et al. 2016; Schiefer and van der Noll 2017). The two components of the index capture complementary attitudinal facets of this construct at different scales. The

[6]Source: The AmericasBarometer by the LAPOP Lab, www.vanderbilt.edu/lapop.

first, trust in the local community, measures proximate, horizontal trust: how respondents view the people immediately around them. The second captures whether respondents name a problem of social disorder or strained social relations—crime, violence, insecurity, gangs, discrimination, or social unrest—as the country's single most serious problem. These problems relate to cohesion because they are breakdowns of these very dimensions: crime, violence, insecurity, and gangs are failures of the peaceful coexistence and shared social order on which cohesion rests; discrimination is a failure of tolerance toward others; and social unrest is open antagonism among groups. These phenomena are precisely those that the literature treats as manifestations of weak cohesion: work on collective efficacy links social cohesion and mutual trust to a community's capacity to maintain social order and limit violence (Sampson, Raudenbush, and Earls 1997), while a long tradition in the study of social capital identifies trust and cooperative norms as foundations of well-functioning communities (Putnam 1993). Moreover, because respondents identify a single most serious problem from a menu that also spans economic, political, infrastructural, and public-service concerns, naming one of these amounts to a comparative judgment that what is most broken in their society is the way people relate to and treat one another, rather than material or institutional conditions. The second component is thus intended to capture whether respondents perceive the social fabric itself as failing, complementing the local, trust-based first component. Nevertheless, the main results of the paper are largely robust to using only this first component: re-estimating the role-model specifications with standardized trust alone leaves the post-match effect and its moderation by red cards stable in sign and magnitude, although more noisily estimated (Appendix Tables A1, A10, A11, and A12).

Other potentially relevant outcomes include attitudes and beliefs about the president, insecurity, and the respondent's economic situation. Finally, I also collect individual characteristics, including region of residence, gender, age, years of education, employment status, civil status, a rural- or urban-residence indicator, and religion.

3.1.2 Football Matches

The dataset on football matches was constructed using online databases from LiveFutbol and the Rec.Sport.Soccer Statistics Foundation.[7] It includes 104 events that took place within one week of a survey round of the AmericasBarometer in their corresponding country. It includes match date and characteristics, including result, number of yellow and red cards, and

[7] Other sources were used to complement missing values, including each country's football-association website and Wikipedia page.

whether the match qualifies as a friendly game, a regular-season game, or an international competition.

These matches represent twenty of the most important football rivalries in Latin America. These rivalries were selected with the objective of working with matches that count as very salient events, at least in the teams' region. All rivalries involve first-division teams with very big fan bases, and they are generally considered the most important rivalries in their countries. The list of rivalries includes, in Argentina, Boca Juniors v. River Plate (Buenos Aires City) and Rosario Central v. Newell's Old Boys (Santa Fe); in Bolivia, Blooming v. Oriente Petrolero (Santa Cruz) and Bolivar v. The Strongest (La Paz); in Brazil, Bahia v. Vitória (Bahia), Corinthians v. Palmeiras (São Paulo), Cruzeiro v. Atlético Mineiro (Minas Gerais), Flamengo v. Vasco da Gama (Rio de Janeiro), and Gremio v. Internacional (Rio Grande do Sul); in Chile, Colo-Colo v. Universidad de Chile (Santiago); in Colombia, América de Cali v. Deportivo Cali (Valle del Cauca), Independiente de Medellín v. Atlético Nacional (Antioquia), and Millonarios v. Santa Fe (Cundinamarca); in Ecuador, Barcelona v. Emelec (Guayas); in Honduras, Motagua v. Olimpia (Francisco Morazán); in Mexico, América v. Chivas (Mexico City-Jalisco) and Monterrey v. Tigres (Nuevo León); in Paraguay, Cerro Porteño v. Olimpia (Asunción); in Perú, Alianza Lima v. Universitario (Lima); and in Uruguay, Nacional v. Peñarol (Montevideo).

3.1.3 Rivalry Salience

The AmericasBarometer does not ask what football team the respondent supports, creating an empirical challenge: I need a measure of each respondent's "exposure risk" to the different rivalries in the sample. I tackle this by leveraging Google Trends data, which provides regional-level (first-level administrative divisions) search intensity for any given search term. I build a measure of regional exposure to each rivalry by retrieving the search intensity of queries of the form <*Team1 Team2*> between March 2015 and March 2020. For example, for the Flamengo v. Vasco da Gama rivalry, I retrieve the search intensity of <*flamengo vasco*>. (When relevant, I use the standard team names abbreviations, in an attempt to emulate the most common search term for each rivalry.) These data allow me to determine which regions are exposed to treatment when a match takes place. To the best of my knowledge, there are no data that link individual attitudes to football team preference, making the region-level analysis the best available alternative.

I normalize the data within each country so that the region most strongly exposed to a given rivalry in this period receives value 100 and all other region-rivalry pairs receive a value

that represents the search intensity relative to the referenced one. I call this the GT-Country Index and exclude from the sample all region-rivalry pairs with a search intensity below 10% of the highest region-rivalry in that country. This threshold has the objective of filtering out regions that are unrelated to a given rivalry and thus should not be considered exposed to matches from that rivalry. For example, Rosario Central v. Newell's Old Boys is a rivalry from the city of Rosario, in the province of Santa Fe, Argentina. The vast majority of followers are located in that province and, indeed, its GT-Country Index scores below 10 in all provinces besides Santa Fe, so respondents from outside Santa Fe will not be considered exposed to matches between these two teams.

In order to make valid comparisons across countries, I need to account for cross-country differences in salience. That is, the search intensity of the highest region-rivalry pair in one country is in general not the same as the search intensity of the highest region-rivalry pair in another country, even though both receive a score of 100 on the GT-Country Index. Thus, I build a new index—referred to as GTI throughout the paper—which adjusts for cross-country differences in salience. Under this index, only one region-rivalry pair in the whole sample receives a value of 100 and all others receive a value that represents the search intensity relative to the reference pair. The region-rivalry pair with highest search intensity in Latin America is Cerro Porteño v. Olimpia in the department of San Pedro, Paraguay. See section A.2 for more details on the sample and main variables used in the analyses.

Working with a treatment defined at the regional level has important implications for interpreting the results. This paper provides evidence on how football matches affect social cohesion in regions where matches are highly salient. I am unable to run any individual-level analyses, such as estimating the impact on football fans v. non-football fans, as I do not observe individual exposure to any given match.

3.2 Identification Strategy

The objective is to identify the causal impact of football matches between traditional rivals on individual attitudes and beliefs. The identifying assumption is that there are no systematic differences between individuals surveyed before and after a match, given a set of fixed effects described below.[8]

The baseline empirical model is the following:

[8]Similar strategies based on the exogeneity of the timing of sporting events have been exploited in the past. See, for example, Dohmen et al. (2006); Healey, Malhotra, and Hyunjung Mo (2010); Depetris-Chauvin, Durante, and Campante (2020).

$$Y_i = \alpha + \beta T^p_{r(i),m(i)} + \Delta' X_i + \Theta_{r(i),m(i)} + \Gamma_{d[m(i)]} + \epsilon_i \quad (1)$$

Where $r(i)$ refers to individual i's region of residence (first-level administrative division), $m(i)$ which match they were exposed to, and $d[m(i)]$ the date of that match. Y_i is any of the outcome variables considered. $T^p_{r(i),m(i)}$ is a treatment indicator taking a value of 1 if the individual was interviewed during the p days after match $m(i)$ in a region $r(i)$ that is in the top half of the GTI distribution for the rivalry represented by that match. X_i is a vector of individual characteristics (gender, age, education, and indicators for living in an urban area, being unemployed, identifying as Catholic, and being single). $\Theta_{r(i),m(i)}$ is a set of region-match fixed effects, imposing that identification comes from variability within each region-match pair. $\Gamma_{d[m(i)]}$ is a set of calendar fixed effects, including day of the week, day of the month, and month of the year, that control for the fact that matches tend to take place on certain days and at certain times of the year. Since treatment varies at the region-match level, standard errors are clustered at that level throughout the paper.[9]

The sample is restricted to individuals interviewed during the p days before and after a match, who were exposed to only one match in that time frame, and who were residing in regions in the top half of the GTI distribution of the corresponding rivalry. Under the baseline specification of $p = 5$, this results in a sample size of 6,118 observations (2,751 treated and 3,367 control) coming from 32 matches and 93 region-match clusters.

As an alternative specification, the binary treatment indicator is replaced with the GTI interacted with a post-match indicator. This corresponds to a treatment-intensity type of regression, where the coefficient of interest represents how much larger is the effect when the salience of the match is one unit larger, as measured by the GTI. This specification has the benefit of using the full sample of respondents and not just those from region-rivalry pairs in the top half of the GTI distribution, resulting in a sample size of around 10,297 observations under $p = 5$ (4,662 treated and 5,635 control) coming from 45 matches and 181 region-match clusters. I also consider larger time windows of $p = 10$, $p = 15$, and $p = 30$.

In Table 1, I provide a balance test where I attempt to predict treatment status. Each column regresses an indicator taking value 1 if the respondent was surveyed within five days after a match on different sets of individual characteristics. To the extent that being interviewed before or after the event is as good as random, individual characteristics should not have predictive power over treatment status. Reassuringly, all coefficients are small and statistically insignificant. Moreover, the F-statistic for joint significance in the last column is

[9]Table A9 in the Appendix shows that results are robust under match-level clustering of standard errors.

0.67 (p=0.73), providing evidence that also the full set of covariates together cannot predict treatment status.

Table 1: Balance table

	Post-Match 5d.							
Female	-0.0012							-0.0013
	(0.0031)							(0.0032)
Age		0.0001						0.0001
		(0.0001)						(0.0002)
Primary education			0.0100					0.0108
			(0.0138)					(0.0141)
Secondary education			0.0059					0.0095
			(0.0149)					(0.0160)
Higher education			-0.0029					0.0020
			(0.0166)					(0.0178)
Urban				-0.0097				-0.0086
				(0.0182)				(0.0182)
Unemployed					0.0013			0.0027
					(0.0083)			(0.0084)
Single						-0.0073		-0.0054
						(0.0052)		(0.0055)
Catholic							-0.0034	-0.0039
							(0.0057)	(0.0056)
Match-region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Calendar FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Control mean	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27
Adj. R2	0.76	0.76	0.76	0.76	0.76	0.76	0.76	0.76
Obs.	10297	10297	10297	10297	10297	10297	10297	10297

Note: All columns regress an indicator for being interviewed within five days after a match on different sets of individual characteristics. The sample includes individuals interviewed within a window of 10 days around a match. Controls include region-wave fixed effects and calendar fixed effects (day of the week, day of the month, and month of the year), to compare treated individuals with their corresponding control group. Standard errors are clustered at the region-match level. * $p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

4 Results

This section presents the results of the paper. First, it introduces the baseline results of how football rivalries affect social attitudes within different time frames; second, it examines how match-level and respondent-level factors moderate the impact of derbies on cohesiveness and what the associated findings imply about the mechanisms at play; and third, it explores alternative outcomes that could also mediate the relationship between football matches and social cohesion.

4.1 Baseline Results

This section shows the effects of derbies on social attitudes and beliefs conducive to social cohesion. The main findings show that cohesiveness improves in the first few days after a match, fading out over a period of 30 days. These effects are robust to various exercises, including a treatment-intensity design, taking each component of the Social Cohesion Index separately, dropping one country at a time from the sample, and adjusting standard errors for multiple hypothesis testing.

Table 2 presents the baseline results. The first column presents the results of the five-day window specification, finding an increase in the SCI of 0.23 standard deviation units during that time period. Columns 2–4 expand the treatment window to 10, 15, and 30 days, respectively, with effects amounting to 0.11σ, 0.09σ, and 0.08σ. Alternatively, columns 5–8 run the treatment-intensity specification, with similar but noisier findings.

Table 2: Main results: Social Cohesion Index

	(1) 5d.	(2) 10d.	(3) 15d.	(4) 30d.	(5) 5d.	(6) 10d.	(7) 15d.	(8) 30d.
Post-Match	0.23**	0.11**	0.09**	0.08**				
	(0.09)	(0.06)	(0.04)	(0.03)				
Post-Match (GTI)					0.0028***	0.0014*	0.0008	0.0016**
					(0.0010)	(0.0008)	(0.0007)	(0.0006)
Match-region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Calendar FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ind. controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Control mean	-0.02	-0.03	-0.04	-0.04	-0.02	-0.03	-0.04	-0.03
Adj. R2	0.09	0.09	0.09	0.09	0.08	0.08	0.08	0.08
Obs.	6118	9311	11478	15390	10297	14889	17838	22645
Clusters	93	105	110	124	181	194	197	209

Note: The outcome variable in all columns is the Social Cohesion Index, a standardized index of trust in the local community and perception of social problems in the country. Post-Match takes a value of 1 if a respondent was interviewed within five days after a match and was located in a region-rivalry cluster with a GTI score in the top half of the distribution. Calendar fixed effects include day of the week, day of the month, and month of the year. Individual controls include a female indicator, age, and indicators for education level, living in an urban area, being unemployed, being single, and identifying as Catholic. Standard errors are clustered at the region-match level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

In the aftermath of a derby—an event that divides communities along the lines of football identities but, by doing so, also promotes bonding across other socioeconomic cleavages—attitudes and beliefs related to social cohesion tend to improve. The magnitude of the effect is comparable to what is found in other studies that analyze the impact of different kinds of events on interpersonal attitudes. For example, Depetris-Chauvin, Durante, and Campante (2020) finds that national football team victories increase trust in fellow countrymen by 14% during the 15 days after the match. Alternatively, Lowe (2021) runs a field experiment where cricket players in India are randomly assigned to playing with or against members of other castes, finding that being teammate with outgroup members increases non-teammate cross-caste friendships by 0.15σ three weeks after the event.

Tables A1 and A2 show that results are similar when considering the two components of the SCI separately, reflecting the broad nature of the impact that derbies have on interpersonal attitudes. In particular, individuals improve their perception of neighborhood as well as national social relations. Moreover, Table A3 indicates that derbies do not

significantly affect the perception of other types of problems in the country. Finally, Figures A6 and A7 show that the baseline results are not driven by any single country in the sample under the binary treatment and the treatment intensity specifications.

In the Appendix, Figures A4 and A5 respectively present the event study plots for the five-day and thirty-day specifications (columns 1 and 4 in Table 2). These plots provide reassuring evidence that there are no pre-trends in social cohesion leading up to a match, which would have posed a threat to the internal validity of the research design. While in the first case there are daily coefficients, in the second case each bin corresponds to five days. Bin zero is the day of the match in the former, and the day of the match plus the four days leading up to it in the latter. Given that the bulk of observations are concentrated in the days near a match, coefficients farther away tend to be less precisely estimated. Any differences across specifications are due to differences in the sample of matches picked up in each case (see Section A.2.3 in the Appendix).

4.2 Heterogeneous Effects

This section presents heterogeneity analyses, exploring whether treatment effects vary depending on characteristics of the match or of the respondent, with the objective of gaining a better understanding of why football matches may affect social attitudes. Table 3 shows how effects change on the basis of match characteristics under the baseline specification. Column 1 considers whether the match ends in a draw, finding little differences in the effect of the match along this line. Although it would be interesting to analyze heterogeneity by winning or losing, the fact that I don't observe what team the respondent supports prevents this possibility. Column 2 focuses on whether the match is of high stakes—an international competition or the knockout stages of a local competition. Although statistical power is low, there is evidence that the effect is considerably larger after high-stake matches, possibly reflecting higher salience among community members.

Table 3: Social Cohesion Index, heterogeneous effects by match characteristics

	(1) Interaction: Draw	(2) Interaction: High-stakes	(3) Interaction: Red cards
Post-Match 5d.	0.21**	0.22**	0.23***
	(0.10)	(0.09)	(0.09)
Post-Match 5d. x Interaction	0.04	0.14	-0.09***
	(0.07)	(0.09)	(0.02)
Match-region FE	Yes	Yes	Yes
Calendar FE	Yes	Yes	Yes
Ind. controls	Yes	Yes	Yes
Control mean	-0.02	-0.02	-0.04
Adj. R2	0.09	0.09	0.09
Obs.	6118	6118	5640

Note: Post-Match takes value 1 if a respondent was interviewed within five days of a match and was located in a region-rivalry pair with a GTI score in the top half of the distribution. A match is considered of high-stakes if it corresponds to an international competition or to the knockout stages of a local competition. In column 3 the interaction variable is de-meaned. Calendar fixed effects include day of the week, day of the month, and month of the year. Individual controls include a female indicator, age, and indicators for education level, living in an urban area, being unemployed, being single, and identifying as Catholic. Standard errors are clustered at the region-match level. * $p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

Finally, column 3 shows that the effect is strongly moderated by the number of red cards awarded during the game.[10] In football as in many other sports, red cards are awarded to players after violent or unethical behavior, causing them to leave the game and often be precluded from future matches. The magnitude of the interaction implies that, in matches with three red cards or more, the effect on social cohesion becomes negative (the average number of red cards is 0.77 in the sample). Table A5 shows similar patterns under the 30-day specification.

When local rivals play violently or unethically against each another, the communities they represent tend to adopt more negative attitudes and beliefs about others. This result aligns with the idea of football matches as spectacles, where players serve as role models for their communities, embodying the possibility of peaceful coexistence with outgroups. The idea that spectacles have the power to influence beliefs has been highlighted since, at least, the work of Debord (1967). Moreover, Depetris-Chauvin, Durante, and Campante (2020)

[10]The number of observations falls slightly because for some matches I could not obtain the number of red cards awarded. Reassuringly, the average effect in this subsample is virtually the same as in the full sample, as observed in the non-interacted coefficient.

also find evidence supporting a role model interpretation of football players. They observe that national football team victories are more likely to improve social attitudes when the ethnic diversity of the roster is higher. That is, when teams tend to incorporate a wider selection of the different groups in their communities, their victories influence more strongly attitudes related to social cohesiveness.

A concern that could jeopardize a causal interpretation of results is that violence in a football match may be endogenous to the state of pre-match social sentiment. In Table A7, I test this by regressing the number of red cards in a match on pre-match SCI among exposed regions, varying the time window between 5 and 30 days. Throughout the table, I fail to reject the null hypothesis that pre-match social cohesion predicts the number of red cards in a game.

Turning to heterogeneous effects across different types of individuals, Tables A4 and A6 show that there are no differences across individuals depending on gender, age, or education. While this is partly to be expected given the widespread interest in football in Latin America, such small differences also imply that football matches do not affect die-hard fans alone, but the community as a whole. This is consistent with the fact that these matches are highly salient events that capture the attention of the vast majority of people in their region of influence, at least for a few days after the event. For example, on the day after the match, it is common that all major newspapers feature in the front page the result of the game and any scandals that may have occurred.

Importantly, in Table A8, I show that baseline results remain statistically significant after adjusting standard errors for multiple hypothesis testing, following Anderson (2008). The table includes the four main regressions from the paper: the 5-day post-match impact on the Social Cohesion Index, and the three heterogeneities by match characteristics (ending in a draw, high stakes status, and number of red cards awarded during the game).

Finally, Table A9 provides evidence that the baseline results remain broadly significant under different approaches to inference, including clustering standard errors at the match level and a randomization inference exercise. The decision of how standard errors should be clustered depends on the level at which treatment is assigned (Abadie et al. 2023); in this setting, treatment is defined by the exposure to a particular match, which varies from region to region, so the relevant treatment cells are region-match pairs. Baseline clustering at this level allows arbitrary correlation among respondents who share the same local exposure. Clustering at the match level provides a more conservative alternative that also allows for correlation across regions exposed to the same match, despite differences in local exposure. Given the small number of resulting clusters, I present wild bootstrap p-values (Cameron,

Gelbach, and Miller 2008). In addition, in the same table I report randomization inference p-values based on 10,000 replications. The average post-match effect remains statistically significant across all four columns under this approach, and Figure A8 shows that the observed baseline estimate lies well on the right tail of the placebo distribution. The interaction with the number of red cards is also significant under this approach (p=0.03), consistent with the match-cluster robust and wild bootstrap results.

4.3 Alternative Outcomes

This section explores alternative outcomes to evaluate other potential mechanisms, including crime, general mood, and attitudes towards the government. Table 4 presents the results. Column 1 shows that the improvement in social cohesion is not driven by changes in perceived insecurity: In the aftermath of a match, respondents do not report a statistically significant change when responding to the question: "Speaking of the neighborhood where you live and thinking of the possibility of being assaulted or robbed, do you feel very safe, somewhat safe, somewhat unsafe, or very unsafe?" The outcome variable is the standardized original response, where higher values indicate greater perceived insecurity.

Inspired by the analyses in Depetris-Chauvin, Durante, and Campante (2020), Column 2 shows that the improvement in social cohesion is not driven (or accompanied) by a generalized sense of optimism or good mood: In the aftermath of a match, respondents are not more likely to think that their own or their country's economic situation is better than twelve months ago. The outcome measure is a standardized index built with the following two questions: (i) "Do you consider that your economic situation is better, the same, or worse than 12 months ago?" and (ii) "Do you consider that the economic situation of your country is better, the same, or worse than 12 months ago?"

Finally, Column 3 shows that the improvement in social cohesion is not driven by, nor results in higher trust in incumbent politicians: In the aftermath of a match, respondents do not report significantly higher trust in the president. The outcome variable is the standardized original response to the trust-in-the-president question, where higher values indicate greater trust. Moreover, this result speaks to the limited effect of popular sport events on the image of incumbents, something that has been debated in the political-economy literature with conflicting results (Healey, Malhotra, and Hyunjung Mo 2010; Fowler and Montagnes 2015; Busby, Druckman, and Fredendall 2017; Corbi 2018).

Table 4: Alternative outcomes

	(1) Feeling Unsafe	(2) Perceived Econ. Sit.	(3) Trust in the President
Post-Match 5d.	-0.03	-0.16*	-0.06
	(0.07)	(0.09)	(0.09)
Match-region FE	Yes	Yes	Yes
Calendar FE	Yes	Yes	Yes
Ind. controls	Yes	Yes	Yes
Control mean	-0.00	0.03	-0.17
Adj. R2	0.08	0.14	0.12
Obs.	6002	6105	6026
Clusters	93	93	93

Note: The outcomes in columns 1 and 3 are standardized measures of the original ordered responses; the outcome in column 2 is a standardized economic-mood index. Post-Match takes a value of 1 if a respondent was interviewed within five days of a match and was located in a region-rivalry pair with a GTI score in the top half of the distribution. Calendar fixed effects include day of the week, day of the month, and month of the year. Individual controls include a female indicator, age, and indicators for education level, living in an urban area, being unemployed, being single, and identifying as Catholic. Standard errors are clustered at the region-match level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

5 Conclusion

This paper investigates the impact of football rivalries on social cohesion in Latin America, leveraging quasi-experimental variation in the timing of football matches and public opinion surveys across eleven countries and twenty rivalries. These rivalries create events that divide and polarize communities from within, yet—in line with Social Identity Theory (Tajfel 1974)—they may also reduce the salience of other, more profound cleavages and offer opportunities for bonding across them (Shayo 2009).

The findings in this study indicate that attitudes conducive to social cohesion tend to improve for up to 30 days after a match, with the strongest effect observed within a 5-day window. Effects are significantly moderated by the number of red cards awarded during the game, suggesting that football players act as role models for their communities. These results are robust to various checks, including alternative model specifications, adjusting standard errors for multiple hypothesis testing, different levels of clustering, randomization inference, testing for pre-trends, and dropping one country at a time from the sample.

However, this study is not without limitations. The nature of the design and the reliance on survey data constrains the analysis to short-term effects and self-reported measures, which may not fully capture the nuances of how football rivalries influence social cohesion. Additionally, the absence of individual-level data on football team preference limits the ability to analyze the differential impact on fans versus non-fans and the relevance of victories and losses. Future research could aim to explore the impact of derbies on behavioral outcomes related to social cohesion, as well as longer-run effects. Moreover, experimental designs that can directly measure individual exposure to football matches would provide more granular insights into the mechanisms at play.

Broader lessons from this study speak to the potential of sports, particularly football, to influence social attitudes and cohesion within communities. Adding to a growing body of work (Depetris-Chauvin, Durante, and Campante 2020; Mousa 2020; Lowe 2021), the findings in this paper suggest that the behavior of athletes on the field can have significant social implications, highlighting the importance of promoting ethical conduct in sports. This policy message should also be read alongside evidence that football can have less benign consequences in other contexts, including increases in interstate conflict around international competitions and short-run increases in violence after unexpected losses (Card and Dahl 2011; Munyo and Rossi 2013; Bertoli 2017). Rather than negating the opportunities documented here, this evidence points to the importance of context and implementation. Governments and civil society organizations could harness post-derby trust surges by

strategically aligning civic initiatives with moments when communities are more predisposed to cooperation, while sports federations could encourage positive role models and emphasize fair play as strategies to sustain the prosocial effects of these events and mitigate social tensions, especially in low-trust environments.

Declaration of generative AI and AI-assisted technologies in the writing process. During the preparation of this work the author used different ChatGPT models to improve clarity and grammar. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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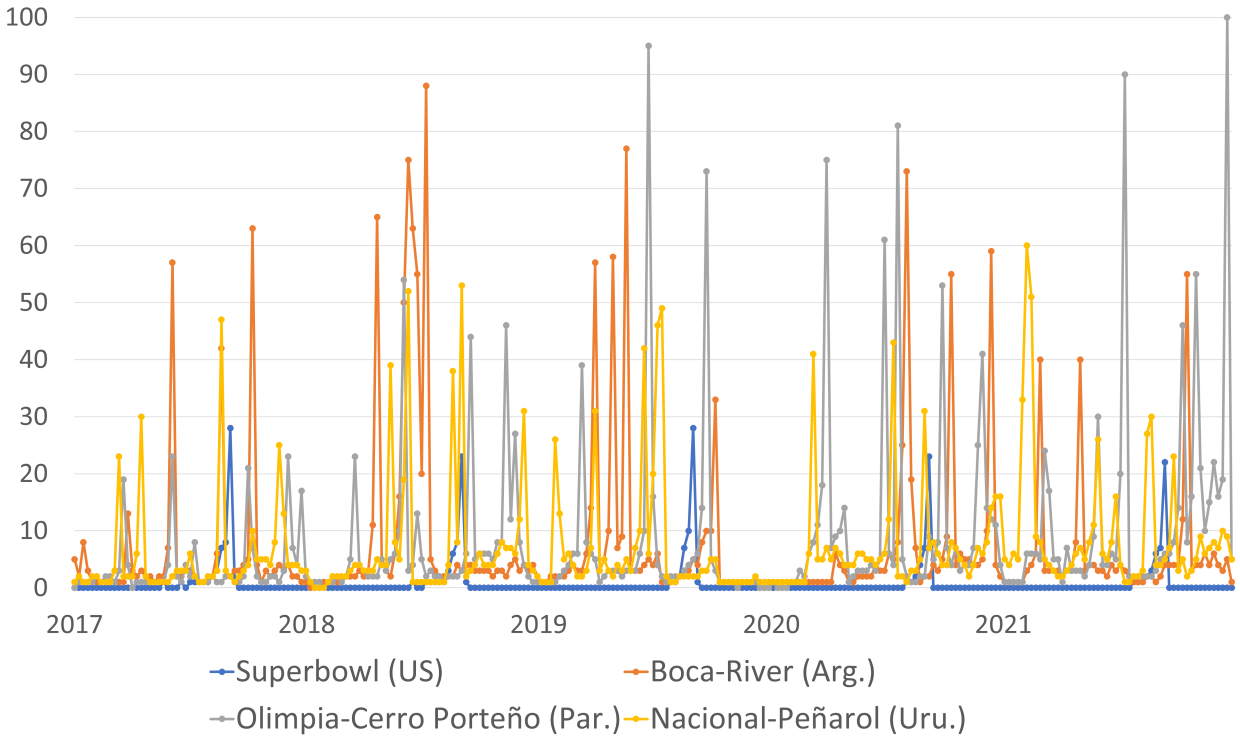
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A Appendix

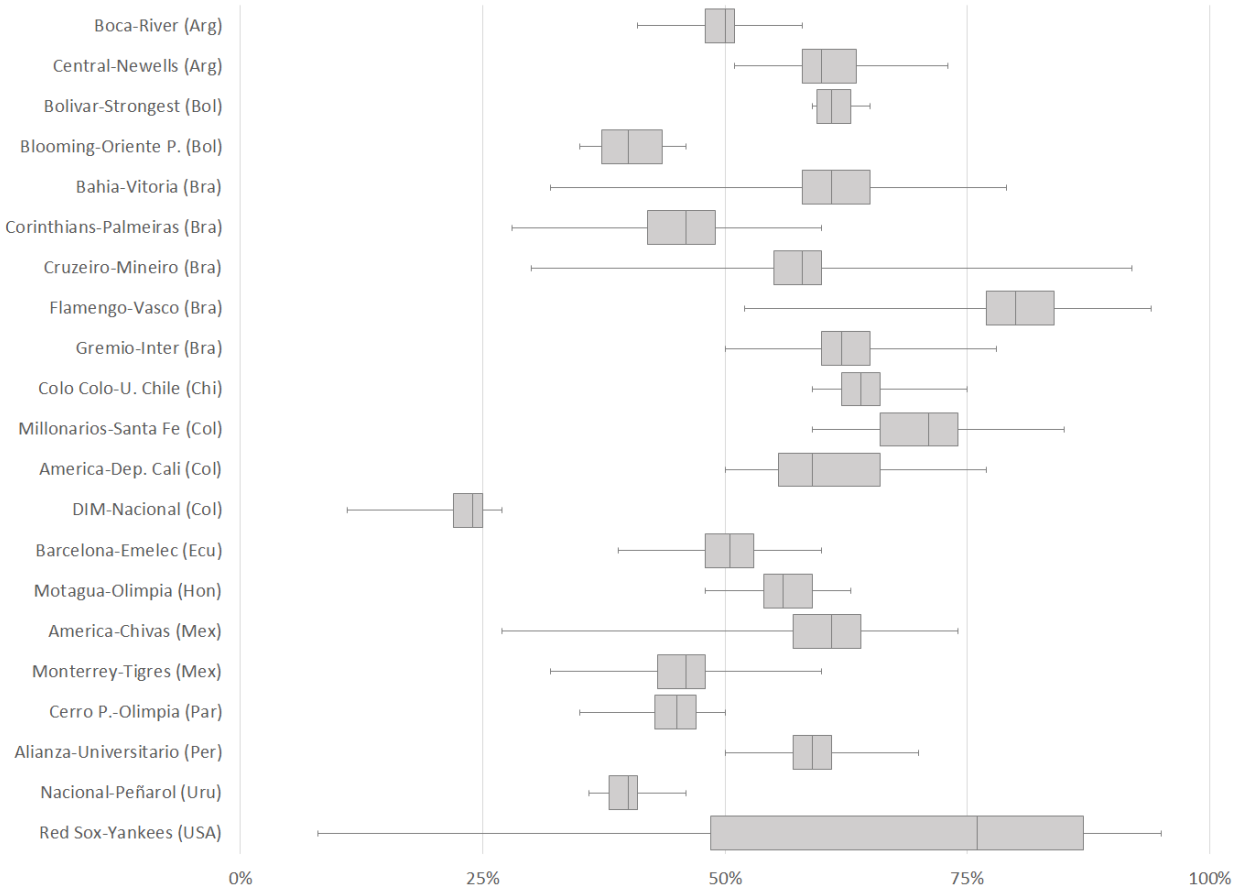
A.1 Background

Figure A1: Salience of selected football rivalries compared to Superbowl in the US



Note: This graph compares Google search intensity between May 2017 and May 2022 for the Superbowl in the United States to the analogous for the main football rivalries in Argentina (Boca Juniors v. River Plate), Paraguay (Olimpia v. Cerro Porteño), and Uruguay (Nacional v. Peñarol), which are some of the most important rivalries in Latin America and have country-wide support. Search intensities were obtained, respectively, under the terms "superbowl", "boca river", "olimpia cerro porteño", "nacional peñarol". Section 3.1.3 provides further details on the Google search intensity data used throughout the paper.

Figure A2: Segregation of followers across cities for each rivalry in the sample



Note: Each box plot represents the distribution of search intensity for the first team in each rivalry pair over total search intensity for both teams across cities within the corresponding country. This measure is a proxy for the distribution of followers of the first team over followers of both teams across cities. For example, an observation with a value of 60% implies that, in that city, the first team in the pair takes 60% of the searches over the total number of searches for the two teams. All rivalries except Red Sox v. Yankees (which is included as reference of a geographically segregated case) tend to have short boxes, implying that cities where one of the teams is relatively more popular do not have a much larger share of followers than cities where the other team is relatively more popular. This is consistent with little geographic segregation.

Figure A3: Distribution of followers across socioeconomic strata

Bolivia

Rivalry	Team	Pop. %	D/E (56%)	C (34%)	A/B (10%)
1	Bolivar	22%	63%	30%	8%
1	The Strongest	9%	50%	38%	16%
2	Blooming	5%	56%	34%	16%
2	Oriente Petrolero	9%	56%	34%	14%

Brazil

Rivalry	Team	Pop. %	Group 1 (49%)	Group 2 (37%)	Group 3 (11%)	Group 4 (4%)
1	Flamengo	17%	56%	30%	10%	4%
1	Vasco	4%	46%	35%	15%	3%
2	Corinthians	14%	45%	44%	8%	3%
2	Palmeiras	6%	42%	39%	13%	6%
3	Gremio	3%	32%	48%	14%	6%
3	Inter	3%	37%	42%	16%	5%
4	Cruzeiro	4%	48%	37%	11%	4%
4	Mineiro	3%	39%	44%	13%	4%
5	Bahia	1%	72%	28%	0%	0%
5	Vitoria	1%	57%	43%	0%	0%

Chile

Rivalry	Team	Pop. %	D (40%)	C3 (29%)	C2 (19%)	ABC1 (13%)
1	Colo Colo	43%	46%	30%	16%	8%
1	U. de Chile	21%	34%	28%	25%	13%

Ecuador

Rivalry	Team	Pop. %	Group 1	Group 2	Group 3
1	Barcelona	30%	29%	21%	22%
1	Emelec	14%	13%	14%	11%
	Ratio	*2.2*	*2.2*	*1.5*	*2.0*

Peru

Rivalry	Team	Pop. %	E (14%)	D (22%)	C (27%)	A/B (15%)	Rural (22%)
1	Alianza Lima	26%	10%	26%	29%	16%	19%
1	Universitario	19%	14%	18%	31%	19%	18%

Note: This figure presents the distribution of football team followers across socioeconomic strata based on country-specific reports from consulting firms.Pop. % indicates the share of respondents that support each team. Each of the columns to the right indicate the share of each team's followers that fall within that socioeconomic stratum (the column title includes the population share in each stratum), except for Ecuador, where this statistic couldn't be computed. Instead, Ecuadorian data shows the share of the population within each stratum that supports each team. Overall, within any given rivalry, we observe a similar distribution of followers across socioeconomic strata. In the case of Ecuador, we observe a stable ratio across strata as in the population. These results are suggestive that there is little socioeconomic segregation of followers within any given rivalry. Bolivian data is based on an in-person survey of 2,000 individuals aged 18–70, during September-October 2019 (Ipsos 2019). Brazilian data is based on an in-person survey of 2,826 individuals aged 16 and over, during January 2018 (Datafolha 2018). Chilean data is based on the average of four in-person survey rounds with between 4,135 and 4,800 individuals aged 16 and over, between 2017 and 2019 (GfK Adimark 2016, 2017, 2018, 2019). Ecuadorian data is based on a survey of 1,900 individuals during October 2018 (Click 2018). Peruvian data is based on an in-person survey of 1,200 individuals aged 18–70, during September 2018 (Datum Internacional 2018).

A.2 Data

A.2.1 Classification of "main problem in the country" responses

I reclassify the 37 topics mentioned as responses to the question of what is the main problem in the country into 6 broad categories. Responses are grouped according to the domain of life to which each problem primarily belongs: *Society* collects problems concerning the horizontal social fabric and the conduct of people toward one another; *Economy*, material and market conditions; *Government*, the vertical citizen–state relationship and politics; *Infrastructure*, physical public goods; *Health & Education*, the corresponding public services; and *Other*, a residual that includes transnational, illicit-market, and catastrophic phenomena that are not primarily about everyday community relations. The resulting categories are as follows:

- **Society**: crime, discrimination, drug addictions, gangs, insecurity, kidnappings, migration, popular protests, violence
- **Economy**: economy, external debt, inequality, inflation, lack of credit, lack of land to farm, poverty, unemployment
- **Government**: bad government, corruption, impunity, politicians, violation of human rights
- **Infrastructure**: housing lack of electricity, lack of water, roads in poor conditions, transportation
- **Health & Education**: education, health services, malnutrition
- **Other**: armed conflict, demographic explosion, drug traffic, environment, forced displacement, terrorism, other

Two boundary cases merit comment, as the line between social and non-social problems can be debated. Drug addiction is classified as a social problem because it is experienced as a community- and household-level ill, whereas drug trafficking is treated separately as an organized-crime and illicit-market phenomenon, frequently transnational and tied to the drug economy. Similarly, discrimination is a horizontal phenomenon among community members and is classified as social, whereas violations of human rights—which in this regional context overwhelmingly denote abuses by state actors such as the police, the military, or the government—are assigned to *Government* as a vertical citizen–state issue.

A.2.2 Football rivalries selection

This section provides a description of the process of selection of football rivalries for the analysis. The key challenge to overcome is that I do not observe what football team the individual supports, which forces me to focus exclusively on rivalries that are very popular in at least one region.

The steps followed were the following:

1. Obtain the list of Latin American countries where football is the most popular sport.

2. Obtain the list of the most popular football rivalries for each country in the sample. For both items 1 and 2, since there is no obvious source to determine these lists, several sources were used: sport magazines articles, interviews to journalists, and Wikipedia pages, among other.

3. Whenever a team showed up more than once (i.e., it had more than one popular rivalry), keep only the main rivalry for that team. This was not common at all – typically teams have only one traditional rival – but in some cases, some sources indicated more than one rivalry. The objective of this step was to avoid including matches that could complicate the interpretation of the coefficients. Examples of rivalries dropped in this step include Chile's *U. de Chile v. U. Católica* (dominated by *Colo-Colo v. U. de Chile*) and Brazil's *Flamengo v. Fluminense* (dominated by *Flamengo v. Vasco da Gama*).

4. Compute the GTI index for each region-rivalry pair (impose zero exposure to foreign rivalries).

5. Drop rivalries that are not highly salient in at least one region. A handful of popular rivalries are actually not highly salient per the GTI in any single region. This happens for two different reasons: (1) when a rivalry is located in a highly populated region and is second in importance to another rivalry in that same region. Thus, even though it has thousands of followers, its GTI is close to zero in all regions of the country. Examples of rivalries dropped in this step include *Independiente v. Racing* (Argentina) and *Estudiantes LP v. Gimnasia LP* (Argentina), which are both located in Buenos Aires. And (2) when a country displays relatively low interest in football compared to the other countries in the sample, so that not even its most popular rivalry in its most popular region receives a reasonably high GTI score. Three rivalries and

their corresponding countries were dropped because of this: *Alajuelense v. Saprissa* (Costa Rica), *Municipal v. Comunicaciones* (Guatemala), and *Caracas v. Táchira* (Venezuela). Thus, even though football is arguably the most popular sport in these countries, the salience of the main rivalry there is low compared to rivalries in the other Latinamerican countries in the sample.

This process resulted in the 20 rivalries included in the sample, belonging to eleven different countries: Argentina, Bolivia, Brazil, Chile, Colombia, Ecuador, Mexico, Paraguay, Peru, and Uruguay. The list of rivalries includes: in Argentina, Boca Juniors v. River Plate (Buenos Aires City) and Rosario Central v. Newell's Old Boys (Santa Fe); in Bolivia, Blooming v. Oriente Petrolero (Santa Cruz) and Bolivar v. The Strongest (La Paz); in Brazil, Bahia v. Vitória (Salvador, Bahia), Corinthians v. Palmeiras (Sao Paulo, SP), Cruzeiro v. Atlético Mineiro (Minas Gerais), Flamengo v. Vasco da Gama (Rio de Janeiro), and Gremio v. Internacional (Rio Grande do Sul); in Chile, Colo-Colo v. Universidad de Chile (Santiago City); in Colombia, América de Cali v. Deportivo Cali (Valle del Cauca), Independiente de Medellín v. Atlético Nacional (Antioquia), and Millonarios v. Santa Fe (Cundinamarca); in Ecuador, Barcelona v. Emelec (Guayas); in Honduras, Motagua v. Olimpia (Francisco Morazán); in Mexico, América v. Chivas (Mexico City-Jalisco) and Monterrey v. Tigres (Nuevo León); in Paraguay, Cerro Porteño v. Olimpia (Asunción City); in Perú, Alianza Lima v. Universitario (Lima); and in Uruguay, Nacional v. Peñarol (Montevideo City).

A.2.3 Measuring rivalry salience with Google Trends data and sample selection

For each rivalry, I define a search term following the form <*Team1 Team2*> and retrieve the regional distribution of search intensity between March 2015 and March 2020. The objective is to mimic as best possible the way people actually run searches on Google, possibly when checking the time of an upcoming match or looking for the outcome of a recent one. For this reason, I use a simplified version of each team's full name. For example, instead of writing <*"Clube de Regatas do Flamengo" "Clube de Regatas Vasco da Gama"*>, I use <*flamengo vasco*>. Importantly, the order of terms is irrelevant. Search intensity refers to the number of queries for the term over the total number of queries in that region and time period. Google reports search intensity normalized with respect to the highest region-period in the sample. Arguably, regions with higher search intensity for a given rivalry tend to be more interested in it. This is confirmed by the distribution of search intensities, as rivalries that are known to be local to a specific region display high search intensity only in those regions.

In the case of countries with multiple salient rivalries, and because different rivalries may attract different volumes of people, a concern is that a score of 100 may imply very different levels of saliency across rivalries. I deal with this by re-normalizing all region-rivalry scores within each country, so that only one region-rivalry attains a score of 100. I call the re-normalized index the GT-Country index. For example, in Argentina, the raw data obtained from Google Trends indicates that Rosario Central v. Newell's Old Boys scores 100 saliency in the province of Santa Fe. But after the re-normalization, and because Boca-River is a much more popular rivalry in that country, the score falls to 11 in that province underthe GT-country index. That is, in the region where it is most salient, rosario Central v. Newell's Old Boys has a search intensity equal to 11% of the search intensity for Boca-River where it is most salient, which is the province of Formosa (and not Buenos Aires City, where Boca Juniors and River Plate are from—likely reflecting the fact that, in relative terms, football is less salient in Buenos Aires than in Formosa). I use the GT-Country index to determine what rivalries are salient in each region. I apply a threshold of 10 on these distributions and assume that respondents below that threshold are not exposed to the rivalry, while respondents above are.

To elicit the intensive margin, and to allow for cross-country comparisons, I need a single measure of relative search intensity across all regions in the sample. I build the Google Trends index—the GTI—by comparing search intensity across the top region-rivalries from each country, obtaining an adjustment factor for each of them. This adjustment factor allows to account for cross-country differences in football rivalry salience. Under the GTI, only one region-rivalry pair receives a score of 100 in the full sample, and all other pairs are normalized with respect to that one. The top three region-rivalry pairs are: (San Pedro, Paraguay - Cerro Porteño v. Olimpia)=100; (Paraguari, Paraguay - Cerro Porteño v. Olimpia)=93; (Formosa, Argentina - Boca Juniors v. River Plate)=90. Finally, note that region-rivalry pairs that combine a region from one country and a rivalry from another are set to 0 in both indices.

A.3 Main Results Appendix

Table A1: Main results: Trust

	(1) 5d.	(2) 10d.	(3) 15d.	(4) 30d.	(5) 5d.	(6) 10d.	(7) 15d.	(8) 30d.
Post-Match	0.185**	0.064	0.048	0.095**				
	(0.081)	(0.059)	(0.052)	(0.044)				
Post-Match (GTI)					0.0015	0.0005	0.0004	0.0015*
					(0.0011)	(0.0009)	(0.0008)	(0.0008)
Match-region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Calendar FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ind. controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Control mean	0.01	-0.03	-0.04	-0.04	-0.05	-0.07	-0.09	-0.08
Adj. R2	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07
Obs.	5957	9061	11178	15007	10054	14527	17413	22143
Clusters	93	105	110	124	181	194	197	209

Note: The outcome in all columns is a standardized (z-score) measure of trust in the local community. Post-Match takes value 1 if respondent was interviewed in the number of days after a match indicated in each column and, in the case of columns 1-4, was located in a region-rivalry pair with GT-Latam index in the top half of the distribution. Calendar fixed effects include day of the week, day of the month, and month of the year. Individual controls include female indicator, age, education level, and indicators for urban status, unemployed status, being single, and identifying as catholic. Standard errors clustered at the region-match level. * $p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

Table A2: Main results: Concerned about societal problems

	(1) 5d.	(2) 10d.	(3) 15d.	(4) 30d.	(5) 5d.	(6) 10d.	(7) 15d.	(8) 30d.
Post-Match	-0.07*	-0.05**	-0.04**	-0.01				
	(0.04)	(0.02)	(0.02)	(0.01)				
Post-Match (GTI)					-0.0012***	-0.0007**	-0.0005*	-0.0005*
					(0.0004)	(0.0004)	(0.0003)	(0.0003)
Match-region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Calendar FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ind. controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Control mean	0.29	0.29	0.30	0.31	0.26	0.28	0.28	0.28
Adj. R2	0.11	0.11	0.11	0.11	0.10	0.10	0.10	0.10
Obs.	6109	9295	11458	15354	10285	14870	17815	22604
Clusters	93	105	110	124	181	194	197	209

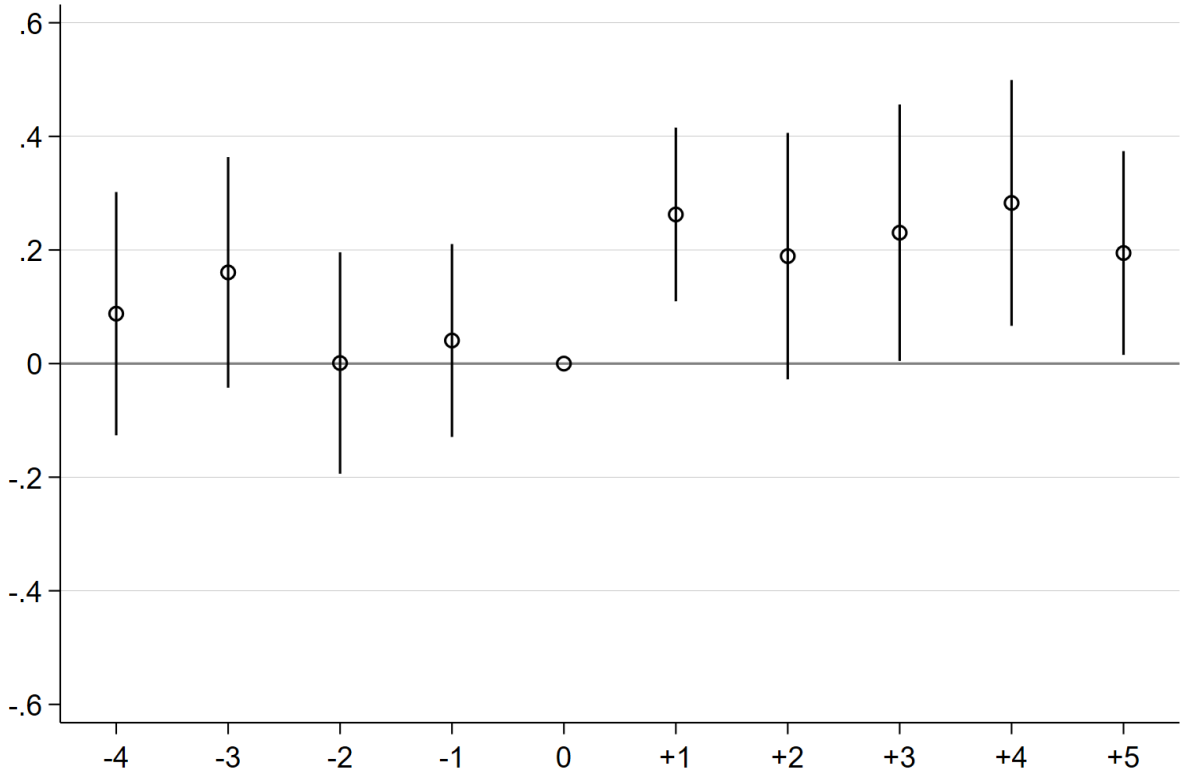
Note: Post-Match takes value 1 if respondent was interviewed in the number of days after a match indicated in each column and, in the case of columns 1-4, was located in a region-rivalry pair with GT-Latam index in the top half of the distribution. Calendar fixed effects include day of the week, day of the month, and month of the year. Individual controls include female indicator, age, education level, and indicators for urban status, unemployed status, being single, and identifying as catholic. Standard errors clustered at the region-match level. * $p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

Table A3: All "main problem" outcomes, 5 day specification

	(1) Society	(2) Economy	(3) Government	(4) Infrastructure	(5) Health&Ed.	(6) Other
Post-Match, 5d.	-0.07*	0.01	0.02	0.01	0.00	0.04
	(0.04)	(0.04)	(0.02)	(0.01)	(0.02)	(0.03)
Match-region FE	Yes	Yes	Yes	Yes	Yes	Yes
Calendar FE	Yes	Yes	Yes	Yes	Yes	Yes
Ind. controls	Yes	Yes	Yes	Yes	Yes	Yes
Control mean	0.29	0.34	0.13	0.02	0.05	0.19
Adj. R2	0.11	0.13	0.07	0.07	0.09	0.21
Obs.	6109	6109	6109	6109	6109	6109
Clusters	93	93	93	93	93	93

Note: Post-Match takes value 1 if respondent was interviewed in the 5 days after a match and was located in a region-rivalry pair with GT-Latam index in the top half of the distribution. Calendar fixed effects include day of the week, day of the month, and month of the year. Individual controls include female indicator, age, education level, and indicators for urban status, unemployed status, being single, and identifying as catholic. Standard errors clustered at the region-match level. * $p < 0.1$, **$p < 0.05$, ***$p < 0.01$

Figure A4: Event-study graph, 5 day specification



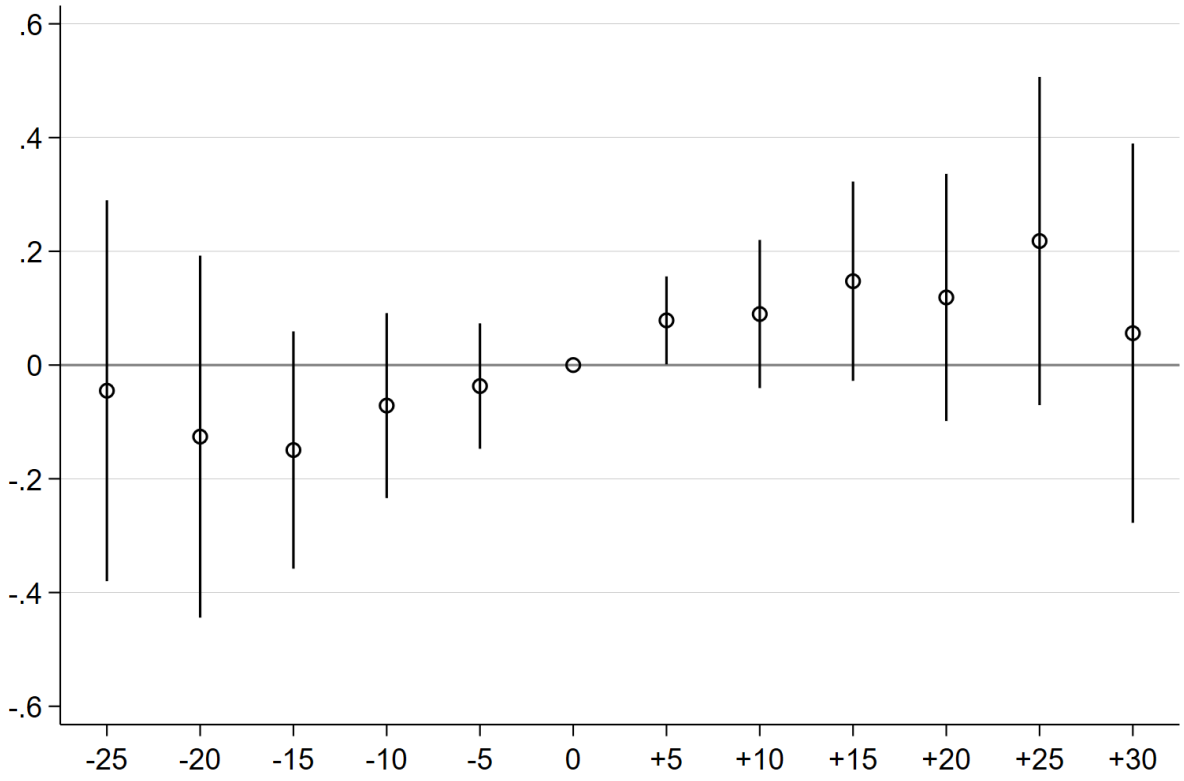
Note: This graph plots the coefficients from regressing the Social Cohesion Index on a set of binary variables indicating one-day periods in the ten-day window around a match (five days before and five days after). Period zero corresponds to the day of the match and is normalized to 0. The full set of controls is included. The confidence intervals are at the 95% level and standard errors are clustered at the region-match level.

Table A4: Social Cohesion Index, heterogeneous effects by individual characteristics

	(1) Interaction: Female	(2) Interaction: Age	(3) Interaction: Educ.
Post-Match 5d.	0.2113**	0.2324***	0.2318***
	(0.0901)	(0.0856)	(0.0861)
Post-Match 5d. x Interaction	0.0373	0.0025*	-0.0025
	(0.0474)	(0.0015)	(0.0049)
Match-region FE	Yes	Yes	Yes
Calendar FE	Yes	Yes	Yes
Ind. controls	Yes	Yes	Yes
Control mean	-0.02	-0.02	-0.02
Adj. R2	0.09	0.09	0.09
Obs.	6118	6118	6118

Note: Post-Match takes value 1 if a respondent was interviewed within five days of a match and was located in a region-rivalry pair with a GTI score in the top half of the distribution. In columns 2 and 3 the interaction variable is de-meaned. Calendar fixed effects include day of the week, day of the month, and month of the year. Individual controls include a female indicator, age, and indicators for education level, living in an urban area,

Figure A5: Event-study graph, 30 day specification



Note: This graph plots the coefficients from regressing the Social Cohesion Index on a set of binary variables indicating five-day periods in the sixty-day window around a match (thirty days before and thirty days after). Period zero corresponds to the day of the match and the four preceding days and is normalized to 0. The full set of controls is included. The confidence intervals are at the 95% level and standard errors are clustered at the region-match level.

Table A5: Social Cohesion Index, heterogeneous effects by match characteristics (30 days)

	(1) I: Female	(2) I: Age	(3) I: Educ.	(4) M: Draw	(5) M: High-stakes	(6) M: Red cards
Post-Match 5d. (GTI)	0.0024**	0.0027***	0.0027***	0.0026**	0.0024**	0.0025**
	(0.0012)	(0.0010)	(0.0010)	(0.0012)	(0.0010)	(0.0011)
Post-Match 5d. (GTI) x Interaction	0.0006	0.0000	-0.0001	0.0003	0.0074***	-0.0008
	(0.0010)	(0.0000)	(0.0001)	(0.0017)	(0.0019)	(0.0007)
Match-region FE	Yes	Yes	Yes	Yes	Yes	Yes
Calendar FE	Yes	Yes	Yes	Yes	Yes	Yes
Ind. controls	Yes	Yes	Yes	Yes	Yes	Yes
Control mean	-0.02	-0.02	-0.02	-0.02	-0.02	-0.04
Adj. R2	0.08	0.08	0.08	0.08	0.08	0.08
Obs.	10297	10297	10297	10297	10297	9603
Clusters	181	181	181	181	181	173

Note: Post-Match takes value 1 if a respondent was interviewed within thirty days of a match and was located in a region-rivalry pair with a GTI score in the top half of the distribution. A match is considered of high-stakes if it corresponds to an international competition or to the knockout stages of a local competition. In column 3 the interaction variable is de-meaned. Calendar fixed effects include day of the week, day of the month, and month of the year. Individual controls include a female indicator, age, and indicators for education level, living in an urban area, being unemployed, being single, and identifying as Catholic. Standard errors are clustered at the region-match level. * $p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

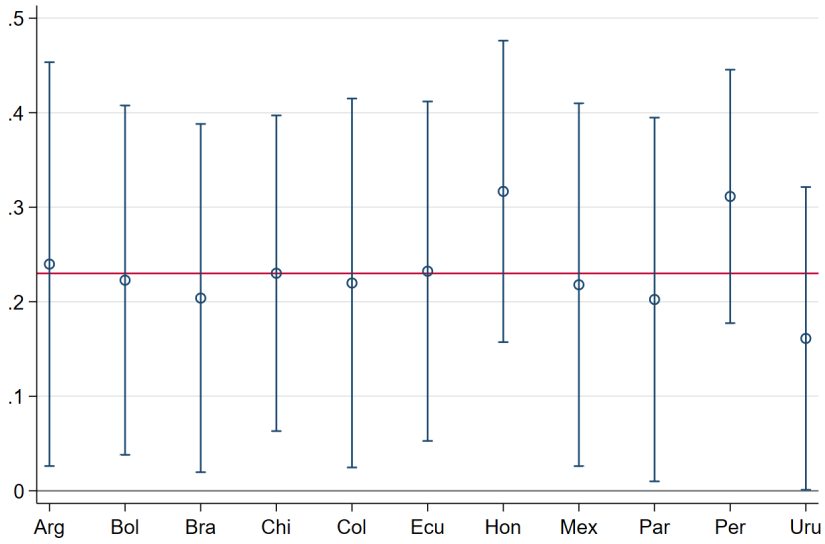
Table A6: Social Cohesion Index, heterogeneous effects by individual characteristics (30 days)

	(1) Interaction: Female	(2) Interaction: Age	(3) Interaction: Educ.
Post-Match 30d.	0.0817**	0.0755**	0.0724**
	(0.0403)	(0.0342)	(0.0338)
Post-Match 30d. x Interaction	-0.0110	0.0014	-0.0052
	(0.0346)	(0.0011)	(0.0039)
Match-region FE	Yes	Yes	Yes
Calendar FE	Yes	Yes	Yes
Ind. controls	Yes	Yes	Yes
Control mean	-0.04	-0.04	-0.04
Adj. R2	0.09	0.09	0.09
Obs.	15390	15390	15390

Note: Post-Match takes value 1 if a respondent was interviewed within thirty days of a match and was located in a region-rivalry pair with a GTI score in the top half of the distribution. In columns 2 and 3 the interaction variable is de-meaned. Calendar fixed effects include day of the week, day of the month, and month of the year. Individual controls include a female indicator, age, and indicators for education level, living in an urban area, being unemployed, being single, and identifying as Catholic. Standard errors are clustered at the region-match level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

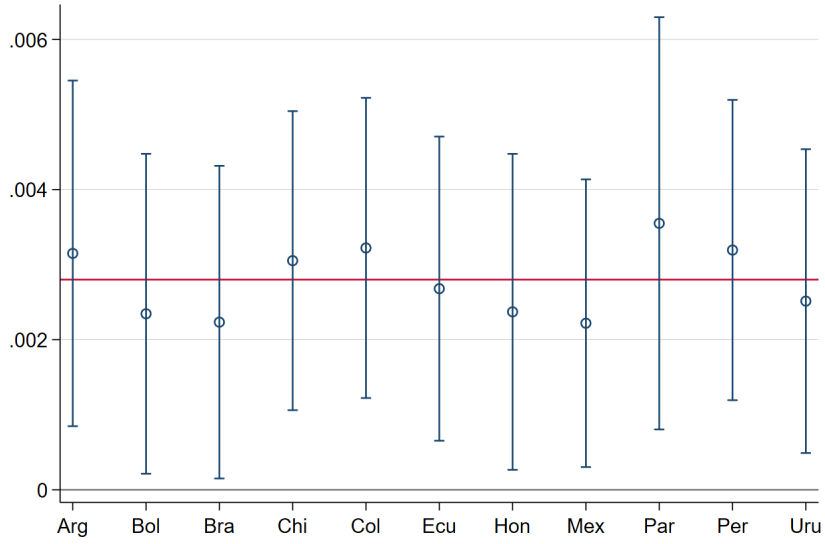
A.4 Robustness checks

Figure A6: Baseline specification, dropping one country at a time



Note: This figure plots the coefficients and 95% confidence intervals from running the baseline specification (5-day binary treatment with full controls) dropping one country at a time. The horizontal red line indicates the effect under the full sample.

Figure A7: Treatment intensity specification, dropping one country at a time



Note: This figure plots the coefficients and 95% confidence intervals from running the 5-day treatment intensity specification with full controls dropping one country at a time. The horizontal red line indicates the effect under the full sample.

Table A7: Pre-match social cohesion does not predict number of red cards

	5 days		10 days		15 days		30 days	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
SCI, Pre-Match	-0.0129	0.0867	0.2460	0.0279	0.1126	-0.1620	0.0852	0.1335
	(0.3235)	(0.4580)	(0.2514)	(0.3443)	(0.3870)	(0.5550)	(0.4898)	(0.7524)
Rivalry FE	No	Yes	No	Yes	No	Yes	No	Yes
Control mean	0.71	0.71	0.69	0.67	0.68	0.68	0.70	0.67
R2	0.00	0.14	0.01	0.18	0.00	0.18	0.00	0.19
Obs.	41	38	45	43	44	41	40	36

Note: Each column regresses the number of red cards in a match on pre-match SCI averages in regions exposed to the match, per the GTI index. Respectively, columns 1, 3, 5, and 7, average over the 5, 10, 15, and 30 days before a match. Columns 2, 4, 6, and 8, present the analogous regressions including rivalry fixed effects. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Baseline results with false discovery rate adjusted standard errors

	(1)	(2) Interaction: Draw	(3) Interaction: High-stakes	(4) Interaction: Red cards
Post-Match 5d.	0.23	0.21	0.22	0.23
	(0.010)**	(0.033)**	(0.015)**	(0.008)***
	[0.027]**	[0.027]**	[0.027]**	[0.027]**
Post-Match 5d. x Interaction		0.04	0.14	-0.09
		(0.520)	(0.104)	(0.000)***
		[0.139]	[0.049]**	[0.027]**
Match-region FE	Yes	Yes	Yes	Yes
Calendar FE	Yes	Yes	Yes	Yes
Ind. controls	Yes	Yes	Yes	Yes
Control mean	-0.02	-0.02	-0.02	-0.04
Adj. R2	0.09	0.09	0.09	0.09
Obs.	6118	6118	6118	5640

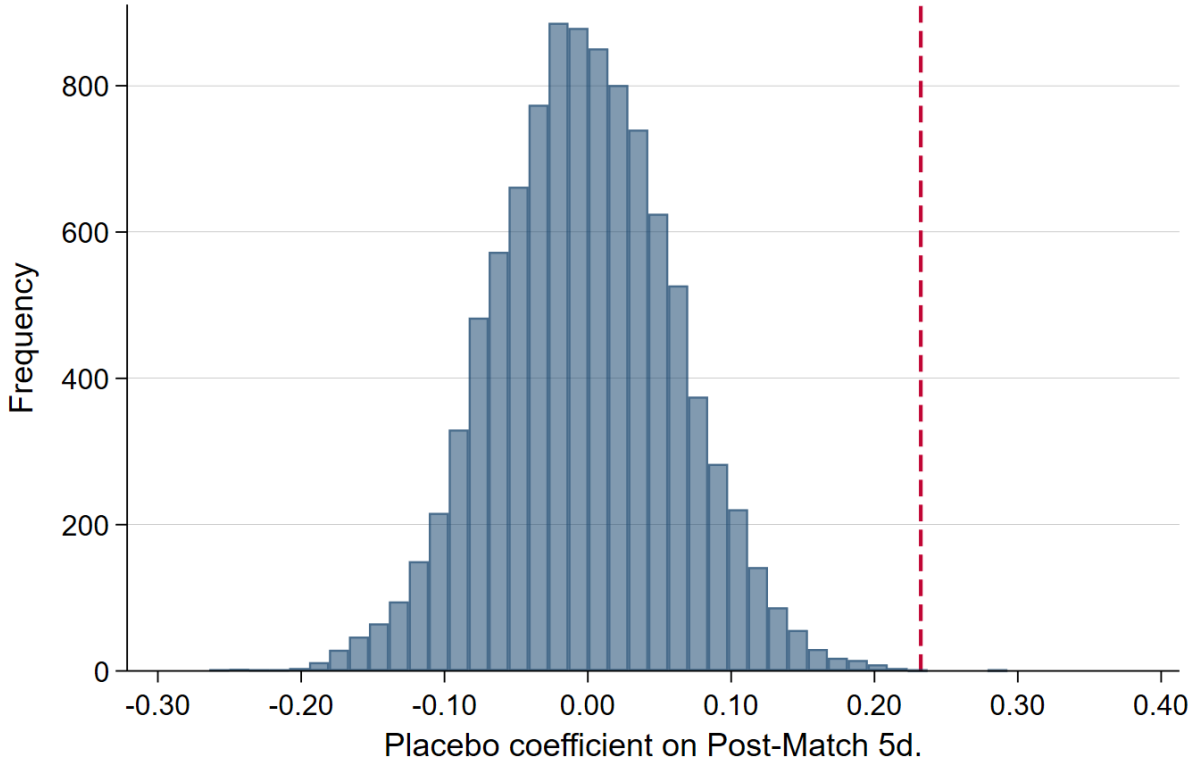
Note: This table presents the baseline specification and the main heterogeneous treatment effects analyses, with robust p-values in parentheses and false discovery rate adjusted p-values in brackets, following Anderson (2008). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: Baseline results with match-cluster, wild-bootstrap, and randomization-inference p-values

	(1)	(2) Interaction: Draw	(3) Interaction: High-stakes	(4) Interaction: Red cards
Post-Match 5d.	0.23	0.21	0.22	0.23
	(0.025)**	(0.057)*	(0.033)**	(0.022)**
	(0.044)**	(0.120)	(0.048)**	(0.060)*
	(0.001)***	(0.007)***	(0.001)***	(0.001)***
Post-Match 5d. x Interaction		0.04	0.14	-0.09
		(0.550)	(0.124)	(0.000)***
		(0.616)	(0.324)	(0.016)**
		(0.577)	(0.220)	(0.031)**
Match-region FE	Yes	Yes	Yes	Yes
Calendar FE	Yes	Yes	Yes	Yes
Ind. controls	Yes	Yes	Yes	Yes
Clusters	32	32	32	30
Control mean	-0.02	-0.02	-0.02	-0.04
Adj. R2	0.09	0.09	0.09	0.09
Obs.	6118	6118	6118	5640

Note: This table presents the baseline specification and the main heterogeneous treatment effects analyses, clustering standard errors at the match level. There are 32 clusters in columns 1–3 and 30 clusters in column 4. For each coefficient, the three rows report match-cluster robust p-values, wild bootstrap p-values (Cameron, Gelbach, and Miller 2008), and randomization-inference p-values (10,000 replications), respectively. The randomization scheme matches the role of each coefficient: For the Post-Match coefficient, each match is assigned a fake date drawn from the empirical distribution of interview dates in regions where its rivalry is salient and treatment is regenerated accordingly, testing the null of no average effect (Borusyak and Hull 2023). For the interaction coefficient, the real match dates are held fixed and the match characteristic is permuted across matches, testing the null of no moderation (Young 2024). In each case the p-value is the share of placebo coefficients at least as large in magnitude as the observed one. Figure A8 plots the placebo distribution of the column-1 coefficient. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Figure A8: Randomization inference: placebo distribution of the baseline 5-day effect



Note: This figure plots the distribution of 10,000 placebo coefficients on Post-Match 5d. obtained under the randomization-inference scheme reported in Table A9. In each replication, every match is assigned a fake date drawn from the empirical distribution of interview dates in the regions where its rivalry is salient, treatment is regenerated from those dates, and the baseline specification is re-estimated. The social cohesion index is held fixed across replications. The dashed vertical line marks the observed estimate.

A.5 Robustness to using the trust component only

Table A10: Trust in the local community, heterogeneous effects by match characteristics (30 days)

	(1) Interaction: Draw	(2) Interaction: High-stakes	(3) Interaction: Red cards
Post-Match 30d.	0.089*	0.095**	0.100**
	(0.050)	(0.044)	(0.043)
Post-Match 30d. x Interaction	0.014	-0.014	-0.069
	(0.057)	(0.126)	(0.043)
Match-region FE	Yes	Yes	Yes
Calendar FE	Yes	Yes	Yes
Ind. controls	Yes	Yes	Yes
Control mean	-0.04	-0.04	-0.03
Adj. R2	0.07	0.07	0.07
Obs.	15007	15007	13415

Note: The outcome is a standardized measure of trust in the local community. Post-Match takes value 1 if a respondent was interviewed within thirty days of a match and was located in a region-rivalry pair with a GTI score in the top half of the distribution. A match is considered of high stakes if it corresponds to an international competition or to the knockout stages of a local competition. In column 3 the interaction variable is de-meaned. Calendar fixed effects include day of the week, day of the month, and month of the year. Individual controls include a female indicator, age, and indicators for education level, living in an urban area, being unemployed, being single, and identifying as Catholic. Standard errors are clustered at the region-match level. * $p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

Table A11: Trust in the local community, baseline and heterogeneous effects with false discovery rate adjusted p-values

	(1)	(2)	(3)	(4)
		Interaction: Draw	Interaction: High-stakes	Interaction: Red cards
Post-Match 5d.	0.185**	0.155*	0.184**	0.209**
	(0.025)	(0.069)	(0.024)	(0.010)
	[0.063]	[0.063]	[0.063]	[0.063]
Post-Match 5d. x Interaction		0.064	0.016	-0.075**
		(0.385)	(0.910)	(0.050)
		[0.148]	[0.352]	[0.063]
Match-region FE	Yes	Yes	Yes	Yes
Calendar FE	Yes	Yes	Yes	Yes
Ind. controls	Yes	Yes	Yes	Yes
Control mean	0.01	0.01	0.01	-0.01
Obs.	5957	5957	5957	5493

Note: The outcome is a standardized measure of trust in the local community. This table presents the baseline specification and the main heterogeneous treatment effects analyses, with robust p-values in parentheses and false discovery rate adjusted p-values in brackets, following Anderson (2008). Calendar fixed effects include day of the week, day of the month, and month of the year. Individual controls include a female indicator, age, and indicators for education level, living in an urban area, being unemployed, being single, and identifying as Catholic. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A12: Trust in the local community, baseline and heterogeneous effects with match-cluster, wild-bootstrap, and randomization-inference p-values

	(1)	(2)	(3)	(4)
		Interaction: Draw	Interaction: High-stakes	Interaction: Red cards
Post-Match 5d.	0.185	0.155	0.184	0.209
	(0.047)**	(0.094)*	(0.047)**	(0.018)**
	(0.084)*	(0.138)	(0.076)*	(0.050)*
	(0.011)**	(0.046)**	(0.011)**	(0.002)***
Post-Match 5d. x Interaction		0.064	0.016	-0.075
		(0.432)	(0.903)	(0.051)*
		(0.502)	(0.900)	(0.098)*
		(0.486)	(0.918)	(0.151)
Match-region FE	Yes	Yes	Yes	Yes
Calendar FE	Yes	Yes	Yes	Yes
Ind. controls	Yes	Yes	Yes	Yes
Clusters	32	32	32	30
Control mean	0.01	0.01	0.01	-0.01
Adj. R2	0.07	0.07	0.07	0.07
Obs.	5957	5957	5957	5493

Note: The outcome is a standardized measure of trust in the local community. This table presents the baseline specification and the main heterogeneous treatment effects analyses, clustering standard errors at the match level. There are 32 clusters in columns 1–3 and 30 clusters in column 4. For each coefficient, the three rows report match-cluster robust p-values, wild bootstrap p-values (Cameron, Gelbach, and Miller 2008), and randomization-inference p-values (10,000 replications), respectively. The randomization scheme matches the role of each coefficient: For the Post-Match coefficient, each match is assigned a fake date drawn from the empirical distribution of interview dates in regions where its rivalry is salient and treatment is regenerated accordingly, testing the null of no average effect (Borusyak and Hull 2023). For the interaction coefficient, the real match dates are held fixed and the match characteristic is permuted across matches, testing the null of no moderation (Young 2024). In each case the p-value is the share of placebo coefficients at least as large in magnitude as the observed one. Calendar fixed effects include day of the week, day of the month, and month of the year. Individual controls include a female indicator, age, and indicators for education level, living in an urban area, being unemployed, being single, and identifying as Catholic. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.